

Work-Energy

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When you throw a rock by hand, it continues to move once released from the hand. The hand doesn't exert a force after the release. **How can we find the speed of the rock?**

In this example Newton's law fail to give an answer.

The rock acquire an energy which has different forms.

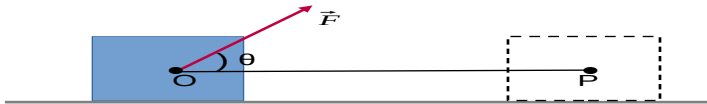
Work of a force

$\vec{F}$ acting on an object of mass m undergoing a displacement $\vec{d}$ parallel to the force is given by:

$$W = \vec{F} \cdot \vec{d} = F \cdot d$$

The unit of work is Newton-meter (N·m) or **Joule (J)**.

When the force is not parallel to the displacement, we consider the parallel component of the force, as illustrated below:



$$W = \vec{F}_{\parallel} \cdot \vec{d} = F \cdot d \cdot \cos(\theta)$$

When the force varies, and we consider a small displacement dl such that the force can be considered constant, the elementary work is given by:

$$dW = F \cdot \cos(\theta) \cdot dl$$

done by the force when this force moves an object from position (1) to position (2) is

$$W_{1 \rightarrow 2} = \int_1^2 F \cdot \cos(\theta) \cdot dl$$

When the force is parallel to the displacement,

$$W_{1 \rightarrow 2} = \int_1^2 F \cdot dl$$

Example 1

When the force is constant the work corresponds to

$$W_{1 \rightarrow 2} = F \int_1^2 dl$$

Exercise 1

Find the work done by the Earth's gravitational force (Weight) on an object of mass m when the object falls from a height h_1 to h_2 (to the ground).

$$W = \int_1^2 \vec{w} \cdot d\vec{y} = - \int_1^2 mg dy$$

$$W = -mgh(y_2 - y_1)$$

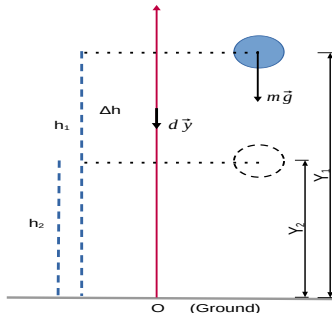
$$\text{but } y_2 - y_1 = -(h_1 - h_2) = -\Delta h$$

So

$$W = mg\Delta h$$

or

$$W = mgh \quad (h_2 = 0)$$



Example 2

The object moves on an inclined plan about an angle α to the horizontal. Frictions are neglected.

$$W = \int_1^2 \vec{w} \cdot d\vec{l} = \int_1^2 mg \cos(\alpha) dl$$

$$W = mg \cos(\alpha) L$$

$$\text{but } L \cos(\alpha) = (h_1 - h_2) = \Delta h$$

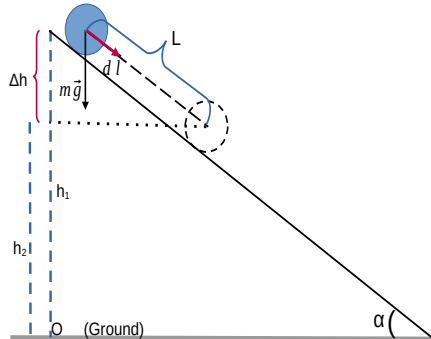
So

$$W = mg \Delta h$$

or

$$W = mgh \quad (h_2 = 0)$$

The same results as free falling object.



The work of the Earth's gravitational force is independent of the path and depends only on the initial and final positions.

Work: positive, negative, or Zero

When an object is shot upward, the work is given by the integral

$$W = \int_1^2 m\vec{g} \cdot d\vec{y} = -mg\Delta h = -mgh \quad (h = 0)$$

The work is negative.

For an object moving along a horizontal line the work of $\vec{w}$ is

$$W_w = \int_1^2 m\vec{g} \cdot d\vec{y} = \int_1^2 mg\cos(\pi/2).dy = 0$$

The work of a force can be positive, negative or zero

Total Work

When there is more than one force exerted on an object, the total work is the algebraic sum of the work done by each force.

$$W = \sum_i^N W_i$$

Kinetic Energy

An external force provides a displacement to an object when a work is done

$$W = \int_1^2 \vec{F} \cdot d\vec{l}$$

The displacement is associated with a change in velocity from v_1 to v_2 . In the case of constant force, the acceleration is constant so,

$$v_2^2 - v_1^2 = 2aL$$

but

$$W = \int_1^2 \vec{F} \cdot d\vec{l} = fL = maL = \frac{m}{2}2aL = \frac{mv_2^2}{2} - \frac{mv_1^2}{2}$$

The work is the difference between the same quantity at different positions, defined as kinetic Energy

$$E_k = \frac{mv^2}{2}$$

Work-Energy Theorem

The work done by an external force on a body equals the change in the body kinetic Energy

$$W = \Delta E_k = \frac{mv_2^2}{2} - \frac{mv_1^2}{2}$$

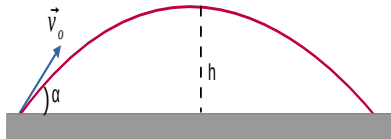
Example

A projectile is shot with an initial velocity of $v_0 = 1\text{ m/s}$ at an angle of 45° with the horizontal. What is the highest point reached by the projectile?

W-E theorem

$$\frac{mv_f^2}{2} - \frac{mv_o^2}{2} = -W_w$$

as $v^2 = v_x^2 + v_y^2$, At the height point $v_f = v_x = v_o \cos(\alpha)$ ($\alpha = 45^\circ$) and $W_w = -mgh$ then, $v_o^2 \cdot \cos(\alpha)^2 - v_o^2 = -2gh$; $h = 2gh / \sin(\alpha) = 27.7 \text{ m}$



Varying force

As $dW = \vec{F} \cdot d\vec{l}$, the work due to displacement under a varying force corresponds to

$$W_{1 \rightarrow 2} = \int_1^2 \vec{F} \cdot d\vec{l} = \int_1^2 m \frac{d\vec{v}}{dt} \cdot d\vec{l} = \int_1^2 m d\vec{v} \cdot \frac{d\vec{l}}{dt} = \int_1^2 \frac{1}{2} m d(\vec{v} \cdot \vec{v})$$

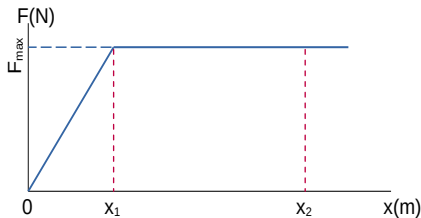
and

$$W = \frac{mv_2^2}{2} - \frac{mv_o^2}{2} = \Delta E_k$$

Example

A body of mass m moves parallel to x-axis. The Force acting on the body is plotted as a function of x (see Figure). Find the change on kinetic energy between x_1 and x_2 .

$$\Delta E_k = F_{max}(x_2 - x_1)$$



$$\Delta E_k = \frac{mv_2^2}{2} - \frac{mv_o^2}{2} = W$$

From the graph:

Kinetic Energy of rotating bodies

To accelerate a particle (object) of mass m from rest to a velocity v , the total work is

$$W = \Delta E_k = \frac{mv^2}{2}$$

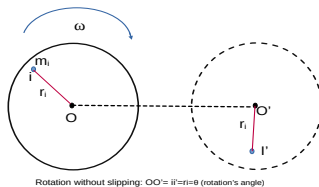
The kinetic Energy is equal to the work done to accelerate the particle from rest to v .

When a rigid body rotates about a fixed axis, the speed of each point i is given by $v_i = r_i \omega$. The kinetic energy corresponding to rotation without slipping is,

$$E_k = \frac{m_i v_i^2}{2} = \frac{m_i r_i^2 \omega^2}{2}$$

The body rotation kinetic energy is

$$E_k = \sum_i \frac{m_i r_i^2 \omega^2}{2} = \left(\sum_i \frac{m_i r_i^2}{2} \right) \omega^2 = \frac{I \omega^2}{2}$$



W-E theorem for rotating bodies

corresponds to

$$W = \frac{I\omega_2^2}{2} - \frac{I\omega_1^2}{2}$$

Example

Find the kinetic energy of the seconds' hand wall watch. The hand is considered as a rod of length $l = 1 \text{ m}$ and of mass $m = 10 \text{ kg}$.

Kinetic Energy

$$E_k = E_{translatio} + I_{rotation} = \frac{I\omega^2}{2}$$

The moment of inertia (we use Huygens' theorem because the rotation axis is different from the symmetric axis) $I = I_{symmetric} + md^2$ (d being the distance between the two axis)

$$I = \frac{ml^2}{12} + m \left(\frac{l}{2} \right)^2 = \frac{ml^2}{4}$$

and for one rotation $\theta = 2\pi$ which takes a time $t = 60 \text{ s}$,

$$\omega = 2\pi/60 = \pi/30 \text{ rd/s}$$

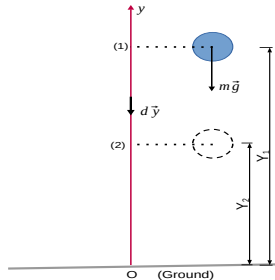
Gravitational potential energy

According to the work-energy theorem, the work of gravitational force is

$$\Delta E_k = W_w = \int_1^2 m \vec{d} \cdot d\vec{y} = mg(y_1 - y_2)$$

$$\frac{mv_2^2}{2} - \frac{mv_1^2}{2} = -mg(y_2 - y_1) = -mgy_2 - (-mgy_1)$$

This is a difference between two energies: $E_p = -mgy$



E_p is the **gravitational potential energy**. In addition we can write, $E_{k2} - E_{k1} = -E_{p2} - (-E_{p1})$ or $E_{k2} + E_{p2} = E_{p1} + E_{k1} = E_{pi} + E_{ki}$ (at any given position) So

$$E = E_k + E_p = \text{Constant}$$

corresponds to **mechanical energy**.

Mechanical energy

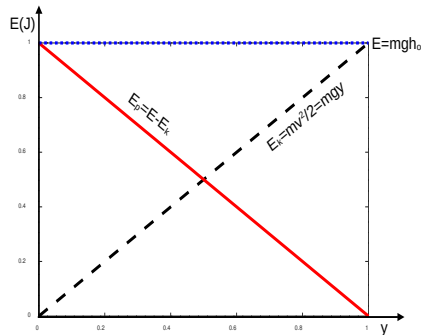
The mechanical energy of an isolated system is conserved

$$E = E_k + E_p = mgh + \frac{mv^2}{2}$$

The energy diagram for free falling object at rest from height h_o , corresponds to

$E = mgh_o = mgh + mv^2/2$. Let $y = h_o - h$ then $E_k = mgy$ and $E_p = mgh = E - E_k$.

The potential energy is found by subtracting the kinetic energy (black dashed curve) from the mechanical energy (blue dotted curve) point by point.



Curved path

The work of the weight is

$$W_w = \vec{w} \cdot \Delta \vec{s}$$

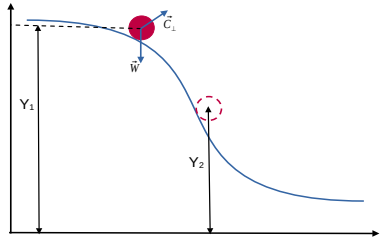
$\Delta \vec{s}$ is the displacement vector.

$$\Delta \vec{s} = \Delta x \vec{i} + \Delta y \vec{j}$$

The work is

$$W = -mg \vec{j} \cdot (\Delta x \vec{i} + \Delta y \vec{j}) = -mg \Delta y$$

$$W = -mg(y_2 - y_1) = -\Delta E_p$$



The gravitational potential energy is $E_p = mgh$, where h is the altitude (vertical height), whatever the path is.

Elastic Potential energy

To move a spring of free length l_o and a constant k from position x_1 to a position x_2 the work corresponds to

$$W = \frac{kx_2^2}{2} - \frac{kx_1^2}{2}$$

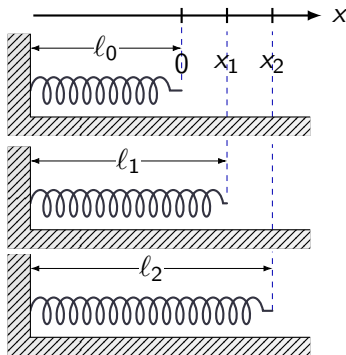
$$x_1 = l_1 - l_o, \quad x_2 = l_2 - l_o$$

The work done by the spring:

$$W_{ela} = -W = -\left(\frac{kx_2^2}{2} - \frac{kx_1^2}{2}\right)$$

$W_{ela} = -(E_{pf} - E_{pi})$ where the elastic potential energy at a given position x corresponding to an elongation (compression) from 0, is

$$E_p = \frac{kx^2}{2}$$

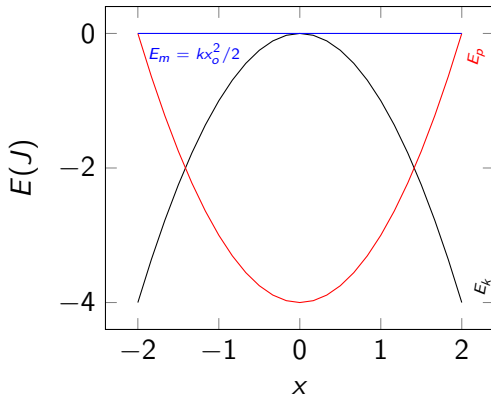


Energy Diagram

If the spring is elongated from rest at a position x_0 then the conservation of mechanical energy reads

$$E_m = \frac{kx_0^2}{2} = \frac{kx^2}{2} + \frac{mv^2}{2} = E_k + E_p$$

and the energy diagram corresponds to,



Conservative Forces

When the work results from a force $\vec{F}$ which produces a displacement $d\vec{l}$, according to the work energy theorem,

$$dE_k = \vec{F} \cdot d\vec{l}$$

When the mechanical energy is conserved, one writes $dE_k = -dE_p$ so,

$$dE_p = -\vec{F} \cdot d\vec{l}; \quad \vec{F} = -\frac{dE_p}{dl} \vec{u}_l; \quad \vec{u}_l = \frac{\vec{l}}{||\vec{l}||}$$

For a vector displacement in three dimensional Cartesian frame;

$$\vec{F} = -\left(\frac{dE_p}{dx} \vec{i} + \frac{dE_p}{dy} \vec{j} + \frac{dE_p}{dz} \vec{k} \right) = -\vec{\nabla} E_p$$

$\vec{F}$ is a conservative force or derive from a potential

Examples of conservative force

- The gravitational force;

$$F = -\frac{dE_p}{dh} = \frac{d(mgh)}{dh} = mg$$

- The elastic force

$$F = -\frac{dE_p}{dx} = \frac{d(kx^2/2)}{dx} = -kx$$

The air resistance force $F_a = -\nu v$ is not conservative because the energy associated to this force

$$W = \int_1^2 -\nu \vec{v} \cdot d\vec{l}$$

depends on the path from position (1) to position (2).

Energy conservation

For an isolated system the total energy (mechanical energy, energy lost by friction, air resistance,...) is conserved.

Example

To one end of a horizontal spring (l_o, k) is attached a mass m which is displaced from equilibrium by x_o . At a given position, neglecting friction, the expression for the total mechanical energy EE is given by:

$$E = \frac{mv^2}{2} + \frac{kx^2}{2} = Constant$$

Taking the derivative of this equation with respect to time leads to:

$$kx\dot{x} + mv \, dx = 0; \quad \ddot{x} + \omega^2 x = 0 \quad \left(v = \dot{x}, \omega^2 = \frac{k}{m} \right)$$

This is a second-order differential equation for x with respect to time t . The solution when $x = x_o$ and v_o at $t_o = 0$, is

$$x = x_o \sin(\omega t + \pi/2)$$